

Towards Fine-Grained Object Manipulation: SAM3-Guided Visuomotor Policy with Persistent Memory Learning and Focused Visual Conditioning

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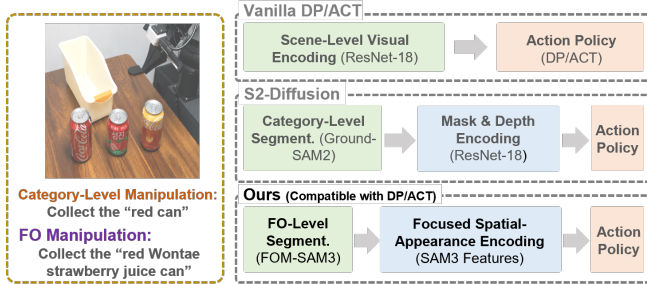


Fig. 1. **Fine-grained target selection for visuomotor manipulation.** (a) An RGB policy learns how to act, but the target identity remains implicit in the scene observation. (b) Mask conditioning provides explicit localization, but assumes that the correct target has already been selected. (c) Our framework retrieves a persistent FO identity, localizes the requested object with frozen SAM3, and provides its focused visual state to the action expert. Thus, changing the retrieved FO Memory changes the target of the learned skill without relearning the skill itself.

Abstract—Imitation learning of visuomotor policies endows the robot with dexterous manipulation skills. However, even in a familiar scene where a robot repeats its tasks, distinguishing a target fine-grained object (FO) from similar objects in the same coarse category remains an unresolved issue. Besides, current visuomotor policies encode the scene image as the visual condition, which might be prone to visual distractors. To address these problems, we first propose FO Memory-enhanced Segment Anything Model 3 (FOM-SAM3). With SAM3 fully frozen, it converts a coarse concept prompt into reusable FO Memory Tokens from a few multi-view registration images. At deployment, the tokens directly prompt frozen SAM3 without registration images or model adaptation. Second, we introduce Focused Spatial-Appearance Encoding (FSAE), which focuses the action policy on ordered target and manipulation-context regions and models their local appearance and relative geometry. FSAE produces a compact condition vector for DP or object condition tokens for ACT without changing either action objective. Experiments on 30 FOs from 4 coarse categories show that FO Memory improves target segmentation, ranks the queried FO above visually similar candidates, suppresses Similar-FO confusion, and rejects absent identities more reliably. Real-robot experiments report completed and failed rollouts and evaluate within-type new-FO reuse with frozen policies.

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I. INTRODUCTION

Visuomotor imitation learning provides a direct route from human demonstrations to closed-loop robot manipulation. Instead of explicitly estimating object poses and designing controllers, the policy learns a mapping from visual observations and robot states to actions. Early work established several important ingredients of this paradigm. Transporter Networks [1] exploit spatial structure by transporting visual features between pick and place locations, showing that strong inductive biases can make manipulation data-efficient. Implicit Behavioral Cloning [2] instead represents the policy as an energy function, allowing multiple plausible continuous actions to be modeled without forcing a unimodal output. As manipulation tasks became more diverse, representation learning became equally important: R3M [3] studies reusable visual representations learned from large amounts of human video, while VIOLA [4] introduces object proposals as priors so that imitation learning can reason over task-relevant regions rather than the entire image. GROOT [5] further moves this idea into object-centric 3D representations. Together, these works have made visuomotor policies increasingly effective at learning manipulation directly from demonstrations, but also reveal a common dependence on how the task-relevant visual state is represented.

A second line of progress concerns the action model itself. Action Chunking with Transformers (ACT) [6] addresses long-horizon imitation by predicting a sequence of future actions as one chunk. This reduces the effective decision horizon and limits the accumulation of step-by-step prediction errors. Diffusion Policy (DP) [7] takes a different approach: it represents the conditional action distribution through diffusion, making it well suited to continuous and multimodal robot behaviors. These ideas have subsequently supported increasingly complex manipulation, including richer 3D-conditioned policies [8] and large-scale contact-rich demonstrations [9]. The important consequence is that modern action experts can already learn *how to execute* many manipulation skills once an informative visual condition is given. However, neither ACT nor DP itself defines which physical object in the scene should provide that condition. Target selection is typically left implicit in the visual observation, demonstration distribution, or task input. This distinction becomes important when the same learned action is valid for several visually similar objects but only one of them is the requested target.

This limitation has motivated increasingly structured visual conditioning for robot manipulation. CLIPort [10] separates semantic “what” from spatial “where” reasoning, while PerAct [11] grounds manipulation in a structured 3D voxel representation. VIOLA [4] uses object proposals to suppress irrelevant image content, and GROOT [5] represents objects explicitly in 3D to improve generalization across scenes and instances. Other approaches incorporate pretrained semantic priors [12], [13], segmentation-aware visual features [14], or language-guided object-centric representations [15]. A particularly relevant example is S^2 -Diffusion [16], which combines promptable semantic information with an explicit spatial representation. By separating category-relevant visual cues from category-irrelevant appearance changes, it allows a skill demonstrated on one object instance to transfer to other instances of the same category. These results establish an important principle: a policy does not need the entire RGB scene if the visual representation can expose the semantic and spatial factors relevant to the skill. Yet category-level generalization and fine-grained target selection are different problems. When multiple Similar-FOs coexist, knowing the category and location of candidate objects still does not specify *which physical instance* the skill should act on.

Promptable segmentation offers a natural way to make object selection explicit. SAM [17] introduced a general interface for segmenting objects from visual prompts, and SAM 2 [18] extended this capability to videos with memory-based temporal propagation. SAM 3 [19] goes further by supporting concept-prompted detection, segmentation, and tracking. A short semantic concept can therefore retrieve all objects that satisfy a description and provide masks and boxes suitable for downstream manipulation. This is powerful, but a semantic concept and a persistent physical-object identity are not equivalent. For example, several objects may all satisfy the concept “can”, while a robot may need to repeatedly retrieve one particular can across different scenes. Recent parameter-efficient adaptation methods show that the prompt interface itself can be specialized. PTSAM [20] adapts SAM to a downstream segmentation task using only a small set of learned parameters and limited annotated images, while M2C [21] searches for a transferable concept embedding directly inside a frozen SAM3 from few-shot masks. These works demonstrate that learned prompts can capture information missing from generic prompts. However, they are designed primarily for segmentation specialization or concept discovery, rather than registering a persistent fine-grained physical-object identity that can directly condition a frozen manipulation skill.

The resulting gap is simple. A visuomotor policy may already know how to grasp, push, or assemble an object, while failure still occurs because the skill is applied to the wrong one. We therefore separate *which object to manipulate* from *how to manipulate it*. We introduce Fine-Grained Object (FO) Memory, which registers a specific physical object as persistent memory tokens in the frozen SAM3 prompt space. FOM-SAM3 retrieves these tokens to localize the requested FO, and Focused Spatial-Appearance Encoding

(FSAE) converts the selected mask, local SAM3 appearance features, and bounding box into the visual condition used by the action policy. DP and ACT retain their original action objectives; changing the retrieved FO Memory changes the target supplied to the learned skill rather than requiring the skill itself to be relearned.

Our contributions are threefold:

- We introduce **FO Memory Learning**, which registers a fine-grained physical object as persistent memory tokens while keeping SAM3 fully frozen. FOM-SAM3 uses the stored memory to retrieve the queried FO, distinguish it from Similar-FOs, and reject the query when the FO is absent.
- We introduce **Focused Spatial-Appearance Encoding (FSAE)**, which converts the selected FO into a policy-ready representation by combining mask-focused SAM3 appearance features with explicit bounding-box spatial state. The same interface supports both DP and ACT without modifying their action objectives.
- We introduce the **FO30 benchmark** for exhaustive FO-present and FO-absent evaluation, and evaluate the complete framework on real-robot manipulation with ordinary and Similar-FO distractors. We further study geometry-compatible new-FO registration while keeping the learned manipulation policies frozen.

II. METHOD

A. Problem Definition

We picture an application paradigm: A robot is deployed in a specific scenario with skillful actions. It can obtain persistent memories of familiar FOs in the scenario, such as “red Wontae strawberry juice can” or “cup lid with a panda handle”. During manipulation, the persistent FO memories are indexed by the FO IDs, then fed to SAM3 as prompt tokens. Thus, the robot can efficiently perceive the memorized FOs without relying on external explicit prompts.

The first problem is the persistent FO memory mechanism built on SAM3. Given an RGB frame I and the memory tokens M_{FO}^i of the i -th FO, FOM-SAM3, denoted as $\mathcal{S}$, can extract the mask Y^i and bounding box B^i , along with the intermediate features H of SAM3, as expressed by,

$$\langle Y^i, B^i, H \rangle = \mathcal{S}(I; M_{FO}^i). \quad (1)$$

When more than one FOs exist, FOM-SAM3 can run in the batch mode, which infers $\{Y^i\}$ and $\{B^i\}$ of multiple FOs according to a batch of FO memories $\{M_{FO}^i\}$ ($i = i_1, i_2, \dots, i_N$). In most cases, an action refers to $N=2$ FOs: one is the active object that the gripper contacts, and the other is the passive object that interacts with the active one.

The second problem is the FO-centric visual conditioning mechanism bridging FOM-SAM3 and the action policy π . Since the preceding FOM-SAM3 has provided exact FO masks and rich visual features, they can be leveraged as the informative inputs of the encoder $\mathcal{E}$ to obtain the visual states of a camera view,

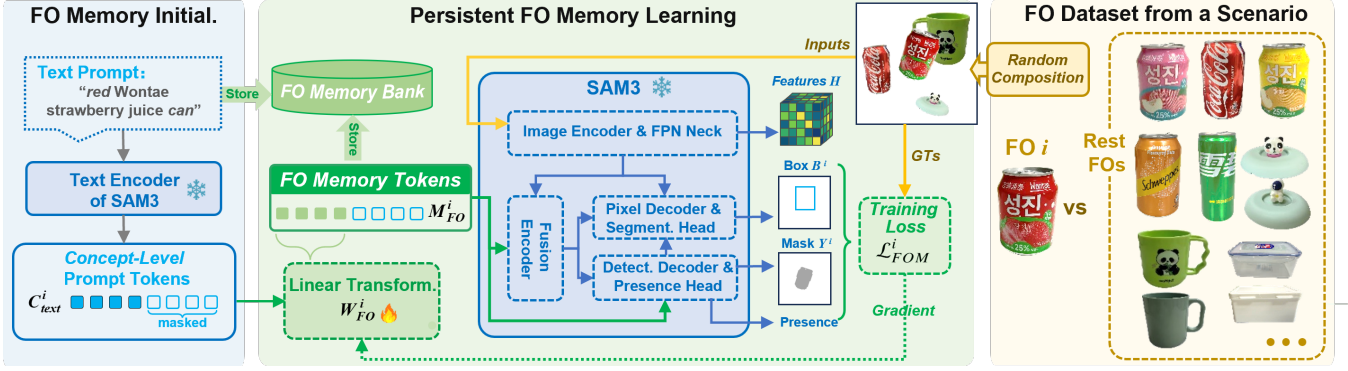


Fig. 2. **Overview of FOM-SAM3 and FSAE.** (a) An FO Prompt retrieves FO Memory Tokens. Frozen SAM3 localizes the target. FSAE conditions DP or ACT. (b) Offline FO Memory Learning updates only valid prompt tokens. SAM3 remains frozen. (c) FSAE uses ROI extraction, mask gating, task context, and relative geometry. It outputs a compact vector for DP or visual tokens for ACT.

$$v^i = \mathcal{E}(H, Y^i, B^i). \quad (2)$$

To enable 3D manipulation, this work adopts a compact dual-view vision setup. The cameras c_1 and c_2 mounted on the left and right arm provide a static third-person view and an active first-person view, respectively. Thus, when two FOs are involved, the action policy is conditioned on four FOs' visual states and the robot state r , as given by,

$$A_t = \pi(v_t^{i_1, c_1}, v_t^{i_2, c_1}, v_t^{i_1, c_2}, v_t^{i_2, c_2}, r_t, z_t), \quad (3)$$

where $A_t \in \mathbb{R}^{T_a \times D_a}$ is the action chunk. $z_t \sim N(0, 1)$ is the noisy action in DP or the style variable in ACT. For brevity, the history input to the policy is omitted in Eq. (3).

B. Persistent FO Memory Learning for SAM3

The original SAM3 is a powerful promptable foundation model, which takes a simple noun phrase or image exemplar as a concept prompt. The concept-level segmentation only provides the masks of coarse-grained objects, such as “green can” or “cup lid”, which cannot effectively guide the FO manipulation. The memory bank of SAM3 in video object tracking stores the transient spatial-temporal tracking memory.

Inspired by the human ability to memorize and recognize familiar FOs, we propose a *persistent FO memory learning* mechanism built upon a frozen SAM3. In essence, the SAM3 inference is conditioned on the internal prompt tokens, which interact extensively with the fusion encoder and decoders of SAM3. The prompt tokens can be temporarily generated from an external text prompt by the pretrained text encoder, or learned from limited data as a persistent memory of a specific FO. Therefore, we define the *persistent FO memory* M_{FO}^i as a set of internal prompt tokens that directly condition the SAM3 modules to segment the memorized FO, while excluding similar FOs belonging to the same concept.

To learn the effective memory of an FO, we first obtain its concept-level prompt tokens $C_{text}^i \in \mathbb{R}^{c \times 256}$ and the text encoder of SAM3 with a noun phrase. Then the memory tokens are formed by a linear transformation of the concept-level prompt tokens, namely,

$$M_{FO}^{i,k} = \begin{cases} W_{FO}^i C_{text}^{i,k}, & \text{if } m_k = 0 \\ C_{text}^{i,k}, & \text{if } m_k = 1 \end{cases} \quad (4)$$

where $W_{FO}^i \in \mathbb{R}^{256 \times 256}$ is a learnable transformation matrix, which is initialized as an identity matrix. m_k is the padding mask that controls the validity of the k -th token.

Since SAM3 already possesses strong generalizable priors on objectness, shapes, and semantics, learning W_{FO}^i mainly aims to distinguish different fine-grained categories within the same concept, such as “red Wontae strawberry juice can” and “red CocaCola soda can”. The learning for each FO is realized in a one-vs-rest (OvR) paradigm. For the i -th FO, while freezing the entire SAM3 and the concept-level tokens C_{text}^i , we only treat the i -th FO as the positive category, and all the rest FOs as negative categories. W_{FO}^i is optimized via gradient backpropagation. The loss function is constructed as,

$$\mathcal{L}_{FOM}^i = \mathcal{L}_{present}^i + \lambda \cdot \mathcal{L}_{absent}^i, \quad (5)$$

where $\mathcal{L}_{present}^i$ is calculated when the i -th FO is present, mainly to guarantee the detection and segmentation accuracy. On the contrary, $\mathcal{L}_{absent}^i$ is calculated the i -th FO is absent, mainly to reject the rest confusing FOs. $\mathcal{L}_{present}^i$ contains all the original loss items used in SAM3 training. $\mathcal{L}_{absent}^i$ contains only the original presence token loss used in SAM3 training. The weight λ is set to 0.1 because the FO-absent image samples are more than the FO-present ones. The token sequence M_{FO}^i given by the learned W_{FO}^i is stored as persistent FO memory in an FO memory bank.

The dataset for FO memory learning contains multiple FOs that appear in a specific scenario, which can be collected in a cumulative manner. Each time a new FO is involved in the current scenario, we simply capture tens of RGB frames of this FO in a pure background from various perspectives. The FO images can be automatically annotated by foreground segmentation. With the single-FO images and their masks, we can further synthesize multi-FO images using mix-up augmentation.

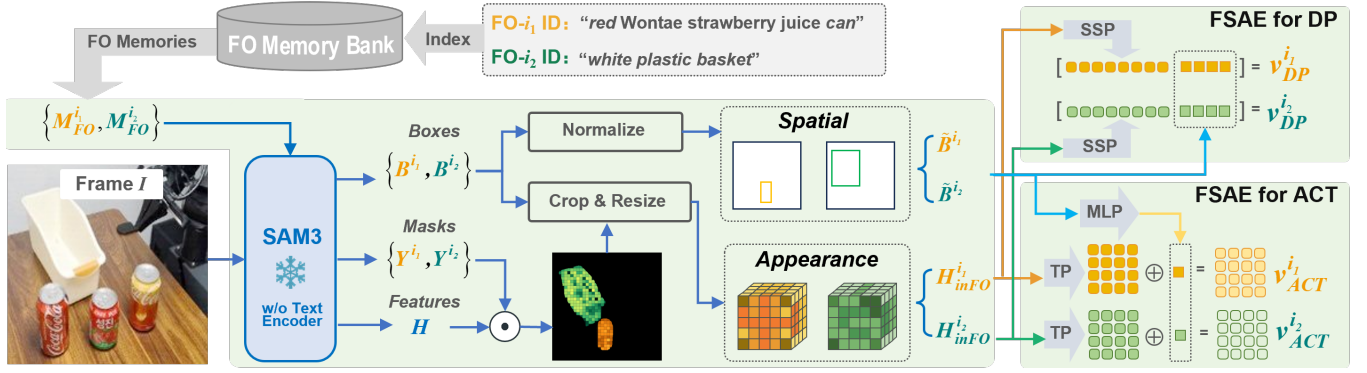


Fig. 3. **Overview of FOM-SAM3 and FSAE.** (a) An FO Prompt retrieves FO Memory Tokens. Frozen SAM3 localizes the target. FSAE conditions DP or ACT. (b) Offline FO Memory Learning updates only valid prompt tokens. SAM3 remains frozen. (c) FSAE uses ROI extraction, mask gating, task context, and relative geometry. It outputs a compact vector for DP or visual tokens for ACT.

C. Focused Spatial-Appearance Encoding

The condition of the visuomotor policy should be focused on target FOs and not distracted by irrelevant objects or background changes. We propose a compact representation that focuses on the spatial and appearance states of target FOs. The spatial state is represented by the bounding box B^i , whose center, size, and aspect ratio form the partial observation of the object pose relative to the camera. The appearance state is represented by the inside-FO features H_{inFO}^i from SAM3's feature pyramid network (FPN), which reflect the pose and interaction information through appearance cues. The inside-FO features are obtained by,

$$H_{inFO}^i = \text{Resize}(\text{Crop}(H \odot Y^i, B^i)), \quad (6)$$

where $\odot$ is the Hadamard product. Namely, we crop the masked FPN feature map within the box B^i , then resize it to a unified resolution. In this work, the unified resolution is 16×16 , while the resolutions of the input image and FPN feature map are 336×336 and 24×24 , respectively. Thus, $H_{inFO}^i \in \mathbb{R}^{16 \times 16 \times 256}$ encodes the appearance cues as an aligned feature map for arbitrarily-shaped FOs.

D. Visual Conditioning for DP and ACT

The proposed FSAE is compatible with the two representative visuomotor policy models DP and ACT. The DP with the CNN structure utilizes a feature vector as the condition. The ACT with the Transformer structure utilizes a sequence of tokens as the condition. (??normalize box)

For DP, we project H_{inFO}^i from 256 dimensions to 16 dimensions as $\tilde{H}_{inFO}^i$, then use the masked spatial softmax (MSS) to obtain a keypoint for each dimension. The resulting 16 keypoints and the bounding box form a visual state, namely,

$$v_{DP}^i = \text{Concat}(\text{MSS}(\tilde{H}_{inFO}^i), B^i). \quad (7)$$

For ACT, we pool H_{inFO}^i to the 8×8 resolution, then project it to 512 dimensions as the appearance embeddings $\{\tilde{H}_{inFO}^{i,k}\}_{k=1:64}$, to align with the token dimension of ACT. Alongside, we project B^i from 4 dimensions to 512 dimensions as the spatial embedding $\tilde{B}^i$. The visual state is

formed by the sum of appearance embeddings and the spatial embedding, as given by,

$$v_{ACT}^i = \{\tilde{H}_{inFO}^{i,k} + \tilde{B}^i\}_{k=1:64}. \quad (8)$$

The execution pipeline of the FOM-SAM3 guided visuomotor policy is summarized below. First, a structured command is sent to the robot, for example, "Action: DP_Collect_CAN, Active FO: red Wontae strawberry juice can, Passive FO: white plastic basket". The IDs of the active and passive FOs are used to index the FO memory tokens from the FO memory bank. Second, the batch of two FO memory tokens is assigned to the prompt token slots of SAM3. If the two FOs are detected by SAM3 in the dual-view images, the segmentation results are used to initialize the video object tracking of SAM3. The tracking result is faster and smoother than frame-by-frame segmentation due to the spatial-temporal continuity. If the tracking fails, the FO memory-driven segmentation is rerun to relocalize the FOs. Third, the segmentation results and the FPN feature map are fed to the FSAE module to obtain the visual states. The dual-view and two-FO visual states, along with the robot state, are used to condition the visuomotor policy to generate a new action chunk.

III. EXPERIMENTS

A. FO Perception Evaluation

1) Dataset and Protocol: We build FO30 with 30 physical FOs from four coarse categories. FO30-S contains 600 standard test images, while FO30-C contains 400 harder images with denser Similar-FO co-occurrence. Each image is paired with all 30 FO prompts, yielding 30,000 prompt tests in total. FO30-S contains 18,000 tests, including 1,268 FO-present and 16,732 FO-absent queries. FO30-C contains 12,000 tests, including 1,192 FO-present and 10,808 FO-absent queries.

2) Baselines and Metrics: We compare SAM3 Text Prompt, Text-Token Optimization [22] (Direct optimization of the K valid 256-D text tokens for each FO; padding and geometry tokens stay frozen), PTSAM [20] (Eight learned 256-D decoder prompt tokens used together with the frozen SAM3 text condition), M2C [21], and FO Memory Learning (Zero-initialized additive residual on the valid post-text-encoder

TABLE I

FO PERCEPTION RESULTS. mIoU IS EVALUATED ON FO-PRESENT QUERIES. RECALL AND PRECISION ARE COMPUTED OVER ALL PRESENT/ABSENT PREDICTIONS, WHILE REJECTION IS EVALUATED ON FO-ABSENT QUERIES AS $1 - \text{FPR}$. ALL VALUES ARE PERCENTAGES.

Method	mIoU $\uparrow$	Recall $\uparrow$	Precision $\uparrow$	Rejection $\uparrow$
<i>FO30-S: Standard</i>				
SAM3 Text Prompt [19]	70.18	73.11	22.61	81.04
Text-Token Opt. [22]	84.77	88.72	28.12	82.81
PTSAM [20]	68.22	71.45	25.58	84.25
M2C [21]	84.85	88.72	28.03	82.74
FO Memory Learning (Ours)	93.04	94.95	51.04	93.10
<i>FO30-C: Challenge</i>				
SAM3 Text Prompt [19]	57.46	62.33	32.29	85.58
Text-Token Opt. [22]	68.84	74.75	38.71	86.94
PTSAM [20]	53.19	59.65	34.25	87.37
M2C [21]	69.63	75.50	38.86	86.90
FO Memory Learning (Ours)	83.25	86.24	56.05	92.54

prompt slots). All methods use the same frozen SAM3, test data, candidate budget, thresholds, and evaluator. We report mIoU on FO-present queries. Recall and Precision are computed over all FO-present/absent predictions. For FO-absent queries, Rejection measures the fraction of correctly rejected queries and is defined as $1 - \text{FPR}$.

All six methods use the maximum budget of 30 epochs or 1,200 optimizer steps. Text-Token Optimization and Concept Embedding Search use a token learning rate of 10^{-3} . FO Memory Learning uses 10^{-4} for their linear or coordinate parameters. Internal Prompt Optimization uses 0.05 for its eight decoder tokens.

3) Results: FO Memory Learning performs best on both test sets. On FO30-S, it achieves 93.04% mIoU, 94.95% Recall, 51.04% Precision, and 93.10% Rejection. The high mIoU and Recall show that the queried FO is accurately localized when present, while the higher Precision and Rejection show that false matches to other objects are substantially reduced.

FO30-C contains denser Similar-FO co-occurrence and is therefore more challenging. Our method still achieves 83.25% mIoU, 86.24% Recall, 56.05% Precision, and 92.54% Rejection. These results show that FO Memory can both retrieve the queried FO when it is present and reject confusing alternatives when it is absent.

4) Prompt-Space Evidence: The trainable prompt routes are compared under the same frozen-SAM3 protocol. Direct token optimization and M2C perform similarly, whereas the FO-specific linear transformation consistently improves localization, retrieval, and rejection on both test sets.

5) Perception Runtime and Storage: We compare SAM3 Text Prompt and preloaded FO Memory Tokens using the same frozen SAM3. After five warm-up passes, 30 measurements give 48.18 ± 4.11 ms for the Text Prompt route and 43.26 ± 5.17 ms for FO Memory. The memory route skips online text encoding and reduces localization latency by 4.92 ms (10.21%).

During deployment, FO Memory is used for global localization and re-localization, while the tracker maintains the selected object between localization calls. FO Memory localization requires 43.26 ms, while tracking requires 40.07 ms.

The 30 stored FO Memory entries occupy only 221.2 KiB in total and persist across episodes.

B. Real-Robot Experiment Setup

1) Platform: We use a single-arm AgileX PiPER robot equipped with two Intel RealSense D405 RGB cameras. One camera is mounted on the wrist for the manipulation view, while the other provides a fixed side view for complementary visual observations. We collect 120 human-teleoperated demonstrations for each task. Demonstrations and policy actions use a 10 Hz time base; the online runner interpolates and publishes robot commands at 50 Hz. Both views are center-cropped to a square and resized to 336×336 . Absolute EE10: 3-D translation, 6-D rotation representation, and one gripper value.

2) Baselines: We choose the 2D image-based Diffusion Policy (RGB-DP) [7] and ACT (RGB-ACT) [6] as standard visuomotor baselines. We also compare with S^2 -Diffusion [16], following its original implementation while adapting it to two-camera mask and depth observations. Ours-DP and Ours-ACT use the same policy architectures, training data, and optimization settings as their corresponding baselines. The only difference is the visual condition.

3) Tasks and Protocol: We design three manipulation tasks: Can Pick & Place, Box Push, and Lid Assembly to study fine-grained target selection when multiple visually similar objects coexist. During training, each task contains only one unique target object and its corresponding task context, allowing the policy to learn the manipulation skill without object ambiguity. During evaluation, we introduce Similar-FOs and visual distractors while keeping the learned policy unchanged. A trial succeeds only when the robot selects the queried FO and completes the full manipulation task. Wrong-FO selection, incorrect interaction, timeout, or incomplete execution are counted as failures. Representative task sequences and evaluation objects are shown in Fig. 4.

Fig. 5 visualizes the ordered perception supplied to the policies throughout each task. Across approach, contact, and object motion, the target and context predictions remain associated with their corresponding mask-gated feature support.

C. FO Manipulation Performance

1) Test Conditions: All policies are trained with a single seen target object for each task. Figure 4 shows the three tasks, seen targets, Similar-FOs, and distractors. All conditions use the same five target markers and ten rollouts per method-task pair.

ID uses the seen target, training background, and task context without extra distractors. The target is moved across the five fixed markers. This setting tests whether each policy learned the basic manipulation skill.

OOD-Distractors changes the background and adds an unrelated object from a different coarse category. The first five rollouts use distractor 1, and the last five use distractor 2. The distractor stays fixed while the seen target moves across the five markers.

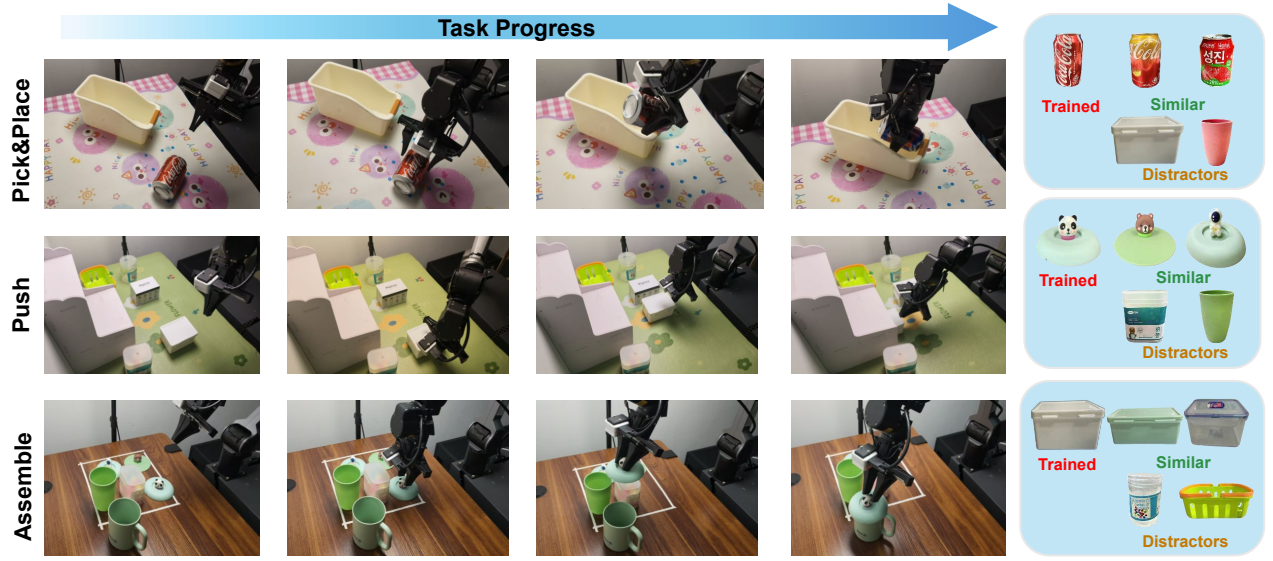
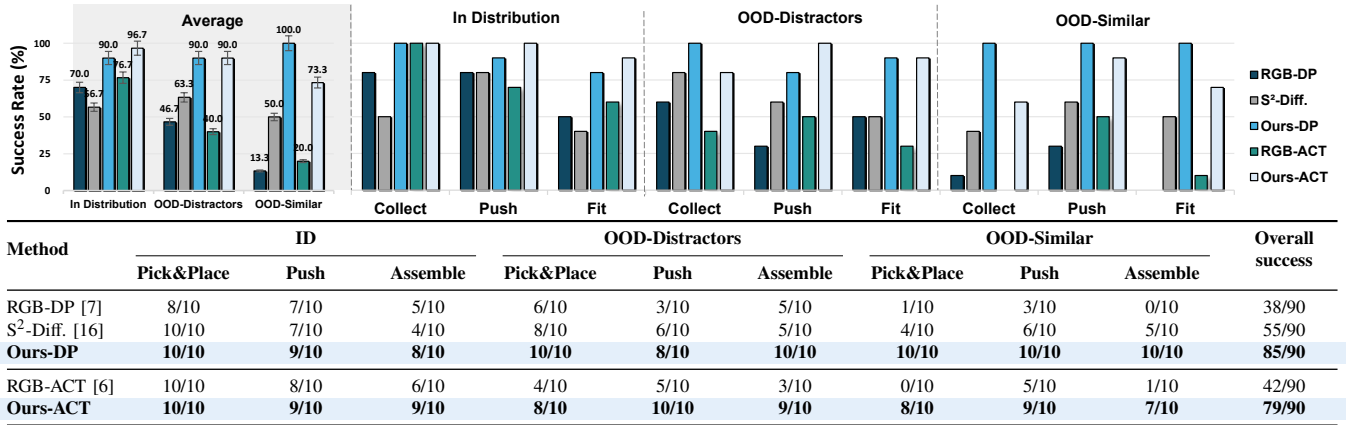


Fig. 4. **Real-robot tasks and evaluation objects.** The rows show Pick & Place, Push, and Assemble. The right column shows the trained FO, a Similar-FO, and task distractors.

TABLE II

REAL-ROBOT MANIPULATION RESULTS. WE COMPARE FIVE METHODS ON THREE TASKS UNDER ID, OOD-DISTRACTORS, AND OOD-SIMILAR CONDITIONS. THE LAST COLUMN AGGREGATES THE 90 ROLLOUTS OF EACH METHOD. OURS-DP AND OURS-ACT GIVE THE STRONGEST OVERALL ROBUSTNESS, ESPECIALLY UNDER OOD-SIMILAR.



OOD-Similar changes the background and places a same-category Similar-FO beside the queried target. The first five rollouts use Similar-FO 1, and the last five use Similar-FO 2. The Similar-FO stays fixed while the queried target moves across the same markers. The robot must select the queried FO and complete the task. A correct action on the wrong object is a failure.

Table II compares all methods under the three test conditions.

The aggregate success rates are 42.2%, 61.1%, 94.4%, 46.7%, and 87.8% for RGB-DP, S²-Diffusion, Ours-DP, RGB-ACT, and Ours-ACT, respectively. Their two-sided 95% Wilson intervals are [32.5, 52.5], [50.8, 70.5], [87.6, 97.6], [36.7, 56.9], and [79.4, 93.0] percentage points. These intervals quantify repeated-rollout uncertainty for the reported trained configurations; they are not confidence intervals over policy-training seeds.

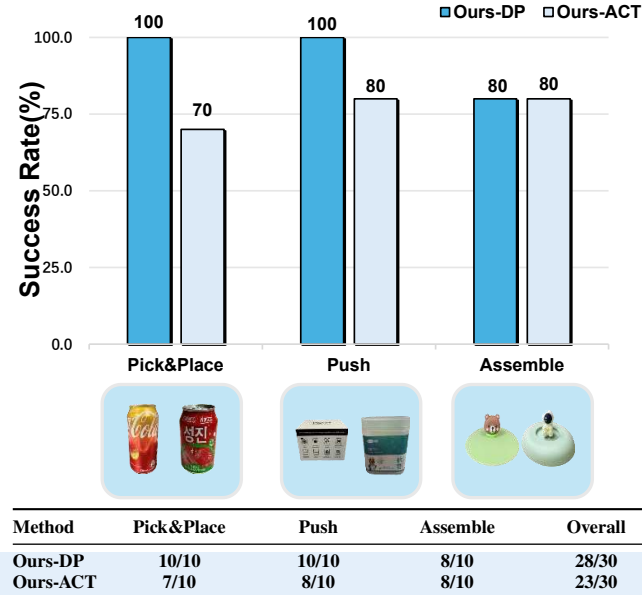
2) Results: All methods retain basic task competence under ID, showing that the corresponding policies learn the required manipulation skills from demonstrations. Their performance separates after visual distractors or Similar-FOs are introduced.

Under OOD-Distractors, the RGB policies remain conditioned on the complete scene and are affected by irrelevant visual content. FSAE instead constructs each FO visual state from the inside-FO SAM3 FPN feature and its bounding box, following Secs. II-C and II-D. This focused spatial-appearance condition reduces the dependence on global background features.

OOD-Similar evaluates the two proposed components jointly. FO Memory retrieves the fine-grained identity that specifies *which object* should be manipulated, while FSAE transforms the corresponding mask, FPN feature, and box into the visual states used by DP or ACT. Ours-DP and

TABLE III

NEW-FO REUSE WITH FROZEN POLICIES. WE CHANGE ONLY THE RETRIEVED FO MEMORY TOKENS AND KEEP FOM-SAM3, FSAE, AND EACH POLICY FROZEN. THE LAST COLUMN AGGREGATES THE 30 ROLLOUTS PER POLICY BACKBONE. BOTH POLICY BACKBONES REUSE THE LEARNED SKILLS ON NEWLY REGISTERED SIMILAR-FOs.



Ours-ACT therefore retain the strongest performance when Similar-FOs coexist with the queried target.

Insight: Fine-grained identity should be grounded before action generation. FO Memory specifies the object identity, whereas FSAE provides the focused spatial and appearance states required by the learned manipulation policy.

1) New FO Reuse with Frozen Policy: We evaluate whether FO Memory enables frozen policies to manipulate newly registered Similar-FOs without additional demonstrations or policy updates. For each newly registered FO, we place it at five different spatial configurations and record the success rate over trials. During evaluation, FOM-SAM3, FSAE, DP, and ACT remain frozen; only the retrieved FO Memory Tokens are replaced according to the queried object. Table III reports this within-type reuse test. The aggregate frozen-policy reuse rates are 93.3% for Ours-DP and 76.7% for Ours-ACT, with 95% Wilson intervals of [78.7,98.2] and [59.1,88.2] percentage points, respectively. Across both frozen backbones, 51 of 60 rollouts succeed (85.0%, 95% Wilson interval [73.9,91.9]). Replacing only FO Memory expands the reachable target set while preserving the learned interaction skill. This is frozen-model, geometry-compatible skill reuse without a new demonstration, not acquisition of a new manipulation skill.

Background Generalization: We evaluate robustness to background variations using the Pick & Place task. The target object, manipulation skill, and policy remain unchanged, while only the background is modified. The evaluation includes one seen background and two unseen backgrounds. Each setting is tested with 10 trials, and success requires

TABLE IV

BACKGROUND GENERALIZATION ON PICK & PLACE. WE COMPARE THREE METHODS ON ONE SEEN AND TWO UNSEEN VISUAL SETTINGS WITH A FIXED TARGET AND POLICY. OURS-DP GIVES THE STRONGEST ROBUSTNESS TO BACKGROUND AND CLUTTER CHANGES.

Method	Seen	Background	Clutter
RGB-DP	8/10	5/10	0/10
Mask-DP	10/10	10/10	2/10
Ours-DP	10/10	10/10	9/10

correctly grasping and placing the target object. The visual settings and success rates are shown in Table IV.

Ours-DP retains the strongest success across the seen, background-shift, and clutter settings. RGB-DP relies on global visual correlations and is sensitive to background variations. Mask-DP reduces background interference through explicit object localization but retains an RGB-derived condition. FSAE instead uses mask-gated SAM3 FPN ROI features to focus on the task-relevant object region and explicit boxes to retain spatial state. The background result is therefore primarily evidence for focused FSAE conditioning, while FO Memory supplies the target identity used to select the region.

D. Limitations

We group the observed failures into four cases. We distinguish FO misidentification, detector miss, tracker drift, temporary occlusion, leaving the field of view, wrong- FO interaction, and manipulation failure after correct identification. Frames with manual correction or re-initialization must be marked separately. **1) Tracking Failure:** The initial localization is correct, but the tracker later loses the target. This occurs under severe occlusion, large viewpoint change, or fast motion. The policy then receives an incomplete or outdated object condition.

2) Perception Failure: FOM-SAM3 may miss the queried FO or select a Similar-FO. This occurs for highly ambiguous objects or under weak registration coverage.

3) Manipulation Failure: The queried FO is detected and tracked, but the policy does not finish the task. Typical causes include grasp error, contact dynamics, poor alignment, and execution drift.

Perception failures include misses and false acceptance of a Similar-FO. Robot failures can occur at localization, tracking, target selection, grasp/contact, transport/alignment, or timeout. The finalized aggregate record does not preserve an audited failure-stage label for every rollout, so no numerical causal decomposition is claimed. Ours-ACT fails 2/10 OOD-Similar Pick & Place trials, and both variants fail 2/10 within-type Assemble trials.

Completed robot cells contain ten rollouts for one trained configuration, not independent policy-training seeds. Exact

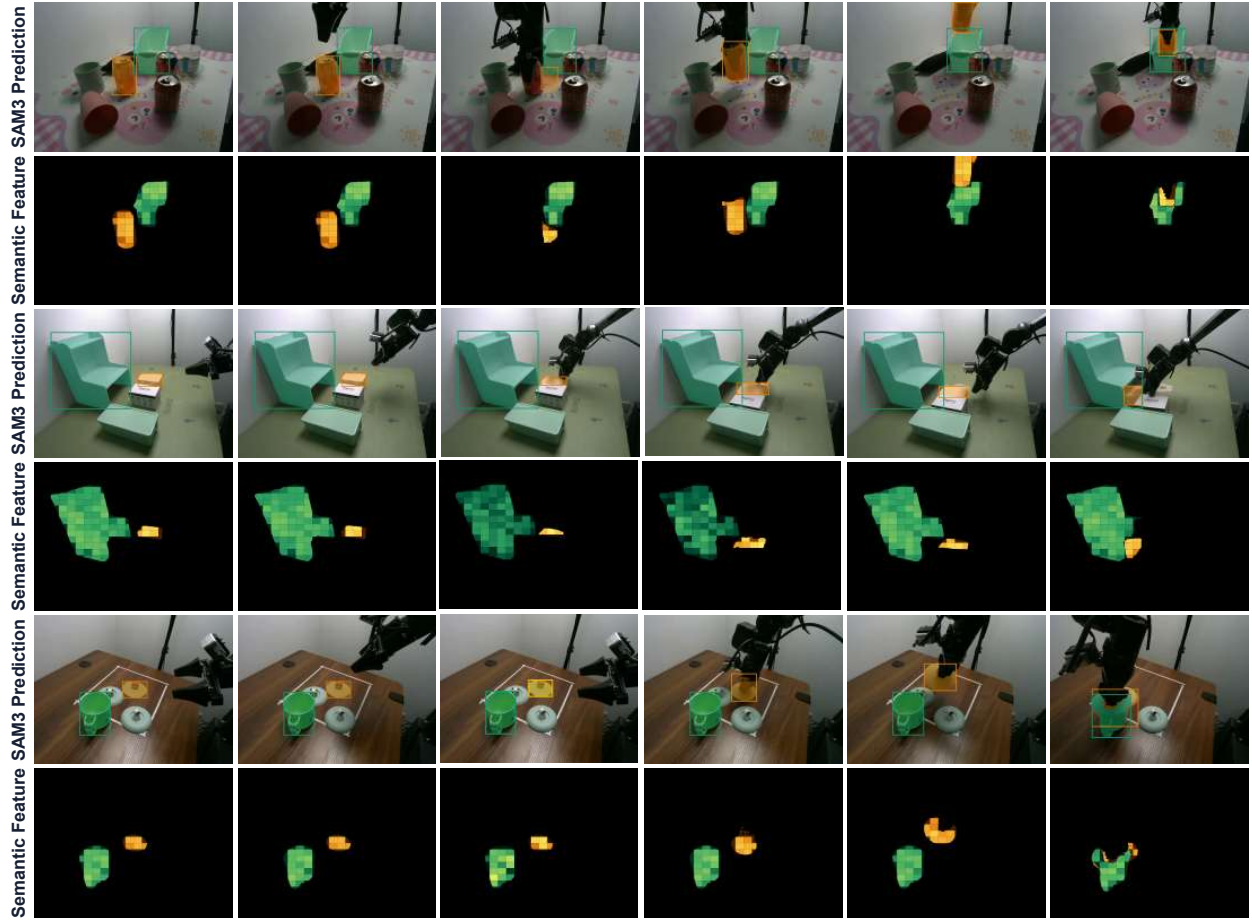


Fig. 5. **Object-centric visual conditions during closed-loop manipulation.** Columns follow representative Pick & Place, Push, and Assembly sequences. SAM3 predictions show the ordered target (orange) and context (green), and the paired semantic maps show the mask-gated features passed to FSAE. The representation preserves task-relevant object support while suppressing most background responses as the scene changes.

task tolerances and timeout values were not retained in the final summary and are not reconstructed. Current trajectory JSONL files lack immutable per-rollout outcome labels sufficient to regenerate every aggregate. No factorial ablation independently isolates mask gating from the explicit box spatial state. Finally, new-FO reuse is bounded by compatible interaction geometry and does not constitute acquisition of a new skill.

IV. CONCLUSION AND LIMITATIONS

This work introduces FOM-SAM3 for persistent fine-grained target memory and FSAE for constructing ordered target/context conditions for DP and ACT. FO30 results

show improved target segmentation, identity ranking, and absent-query rejection with frozen SAM3. Completed real-robot evaluations show stronger robustness to ordinary distractors and Similar-FOs, and within-type tests demonstrate query switching among newly registered FOs without policy retraining. The evidence remains limited by ten-rollout aggregates, absent cross-seed policy statistics, and the lack of matched FSAE component ablations; the reported reuse result therefore applies only to the tested within-type, geometry-compatible objects.

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APPENDIX I

FO MEMORY LEARNING IMPLEMENTATION

TABLE V
FO MEMORY LEARNING IMPLEMENTATION. THE TABLE SUMMARIZES THE TRAINING PROTOCOL AND FROZEN COMPONENTS.

Component	Setting
Frozen modules	SAM3 image encoder, text encoder, fusion encoder, mask/box decoder, and native presence head.
Prompt layout	32 text slots of dimension 256 plus one frozen geometry token; only target-FO valid text slots are changed.
Trainable state	One identity-initialized, bias-free 256×256 linear operator W_i per FO.
Online epoch	320 scenes: 256 target-present and 64 target-absent scenes containing a hard similar object; 2–4 objects per scene.
Budget	Effective batch 8, 40 optimizer steps/epoch, at most 30 epochs or 1,200 steps.
Optimizer	AdamW; linear-operator LR 10^{-4} , weight decay 0.1, gradient clip 1.0, bfloat16.
Schedule	One inverse-square-root schedule with 20-step warmup, 20-step timescale, and 20-step cooldown.
Validation	Fresh 320-scene validation every 5 epochs; early stopping from epoch 10 with patience 2 and minimum improvement 0.005.
Selection	Mean of Joint-pass, Rank@1, 1–raw Similar-FO FPR, 1–Absent FPR, and positive F1; oracle-IoU guard 0.95.

Positive candidates use SAM3’s official Hungarian matcher with class, box-L1, and GIoU costs 2, 5, and 2, and focal parameters $\alpha = 0.25, \gamma = 2$. Positive mask supervision combines 0.8 Dice and 0.2 BCE; box-L1 and GIoU weights are both 0.1. Positive and hard-negative presence means are 1.0 and 0.1. The candidate-to- presence gradient is detached, while the native SAM3 presence head itself stays frozen.

Every online epoch is regenerated from registration RGB/mask pairs under a fixed random contract shared by all methods. The 256 positive scenes contain the target, a configured similar FO, and, when at least three objects are sampled, an ordinary non-similar distractor. The 64 hard-negative scenes remove the queried target while retaining a configured similar FO. Photometric augmentation uses mild brightness, contrast, or gamma variation; blur, noise, or JPEG degradation is applied sparsely. Global crop and rotation are disabled so that the accepted target area and visibility constraints are controlled by the scene compositor. At least 95% of each supervised target mask remains visible, and overlap occlusion is capped at 5%. Test images and masks are explicitly forbidden as compositor material.

For each positive pair, the matcher assigns the queried FO’s GT mask and tight box to one of SAM3’s candidates. The loss combines mask Dice and BCE, object classification, box L1/GIoU, and positive presence. For a target-absent hard negative, only the native presence pathway is supervised; assigning the visible similar object as a negative mask target would incorrectly teach its geometry to the queried identity.

A. Detection Score and Acceptance Rule

For one image–FO query, frozen SAM3 returns 200 raw mask/box candidates, one native presence logit, and an unfused object logit per candidate. Candidate q is ranked by

$$s_q^{\text{joint}} = \sigma(s_q^{\text{pres}}) \sigma(s_q^{\text{obj}}). \quad (9)$$

A candidate is accepted when $s_q^{\text{joint}} \geq 0.5$; its mask is binarized at 0.5, and mask NMS uses IoU 0.8. The same rule, candidate budget, thresholds, and evaluator are applied to all five perception methods. Rank@1 is computed before thresholding and NMS so that a low-confidence landing on a similar object cannot disappear from the identity-ranking diagnostic.

B. Policy-Specific Visual Conditioning

For DP, the mask-gated ROI is projected to 16 channels and reduced by masked SpatialSoftmax to 16 two-dimensional keypoints. Concatenating the resulting 32-D appearance vector with box4 yields 36 dimensions per FO and 144 dimensions for two cameras and two FO slots. For ACT, each ROI is pooled to an 8×8 grid and projected to 512-D tokens. The projected box spatial state is added to each of the 64 appearance tokens, yielding 256 tokens for two cameras and two FO slots.

SAM3 remains outside the policy optimization graph: its cached FPN2 features, masks, and boxes are fixed inputs. Gradients from the action loss update the trainable FSAE components and the action backbone, but not SAM3.

Table VI compares the DP and ACT interfaces.

TABLE VI
FSAE INTERFACES FOR DP AND ACT. THE TWO INTERFACES USE THE SAME FROZEN SAM3 OUTPUTS.

Item	Ours-DP	Ours-ACT
ROI output	16 heatmaps and masked SpatialSoftmax; 32-D appearance plus box4 per FO	Mask-gated 8×8 grid; 64 tokens per FO, projected from 256-D to 512-D
Two cameras, two slots	144-D visual condition	256 visual tokens of dimension 512
Policy input	Two EE10 states give 164-D observation condition; plus 128-D time embedding gives 292-D FiLM condition	Visual tokens plus latent and state tokens give 258 encoder tokens
Action supervision	Diffusion denoising MSE; gradients update FSAE, SpatialSoftmax adapter, and DP	Action L1 plus VAE KL; gradients update FSAE, token projections/embeddings, and ACT
Training budget	Batch 16, 150k steps, prediction horizon 24, action horizon 16	Batch 16, 100k steps, chunk 30, temporal-ensemble coefficient 0.01
Configuration family	FOM-SAM3 + FSAE with DP	FOM-SAM3 + FSAE with ACT

Table VII lists the policy configurations.

TABLE VII
POLICY TRAINING AND INFERENCE CONFIGURATION. THE BATCH SIZE IS 16 FOR EVERY TASK AND BACKBONE.

Item	Ours-DP	Ours-ACT
Optimizer	Adam, LR 10^{-4} , betas (0.95, 0.999), weight decay 10^{-6}	AdamW, LR 10^{-5} , weight decay 10^{-4}
Schedule	Cosine decay; 500-step warmup	Constant optimizer settings
Training budget	Batch 16; 150k updates	Batch 16; 100k updates
Temporal input	Current image; EE10 at $t - 1, t$	Current image and current EE10
Action target	24 steps, $t - 1:t + 22$; execute $t:t + 15$	30 steps, $t:t + 29$; emit one action with temporal ensembling
Normalization	Visual identity; state/action min-max	Visual identity; state/action mean-standard deviation
Action model	1-D conditional U-Net, widths 512/1024/2048, kernel 5, group norm 8, scale-and-bias FiLM	Transformer width 512, 8 heads, FFN 3200, 4 encoder and 1 decoder layers
Stochastic objective	DDPM, 100 train and 100 checkpoint inference steps, squared-cosine schedule, epsilon prediction	32-D VAE latent, 4 VAE encoder layers, KL weight 10
Deployment	16-action receding-horizon chunk	Temporal-ensemble coefficient 0.01; one action/control step

RGB and FSAE variants use identical demonstrations, state/action representations, action objectives, and training budgets. At inference, FSAE and the policy are frozen together; registering a new FO changes only the FO Memory Tokens stored in the bank.

C. Baseline Fairness and Policy Equivalence

The saved checkpoint configurations confirm that each RGB-ours pair changes the visual condition while retaining the corresponding action-policy formulation. RGB-DP and Ours-DP use the same 10-D state/action representation, conditional 1-D U-Net, 24-step prediction horizon, 16-step action horizon, optimizer, batch size 16, and 150k-update budget. RGB-ACT and Ours-ACT use the same 30-step chunk, transformer/VAE action model, optimizer, batch size 16, 100k-update budget, and temporal-ensemble coefficient. Thus, the DP denoising objective and ACT action/VAE objectives are unchanged; only the RGB visual encoder is replaced by the FSAE condition interface. All methods use the same two-camera robot observations and demonstrations.

S²-Diffusion is reproduced in the same DP training and rollout harness. Its two channels per camera are the union foreground mask and monocular depth, matching the baseline’s semantic-spatial conditioning intent. It shares the DP backbone, EE10 state/action, 24/16 horizons, optimizer, batch size 16, 150k updates, and rollout protocol with RGB-DP and Ours-DP. It does not use FO Memory or FSAE, and the main table reports only this depth-enabled variant. This isolates the comparison to visual conditioning without giving our method an additional action-model or training-budget advantage.

D. Online Detection and Tracking

At episode reset, FOM-SAM3 retrieves the required FO Memory Tokens from the bank and supplies them to the frozen 504-pixel SAM3.1 multiplex detector, which must initialize every ordered slot. The detected mask and box provide a global visual identity anchor for the requested FO. After resizing, these masks seed independent 336-pixel trackers for the two

camera streams. Each tracker memory bank uses the recent image/mask history and its previous state to propagate local motion and maintain the selected instance between detector calls.

Normal control runs the tracker, FPN2, FSAE, and DP/ACT. If a required slot is invalid, actuation stops and FOM-SAM3 repeats detection with the retrieved FO Memory Tokens before the tracker is re-initialized. The persistent FO memory bank therefore supports long-term identity retrieval and global re-localization, whereas tracker memory performs short-term temporal maintenance only. Tracker state is camera-specific, episode-local, never changes slot order, and never updates the FO Memory Tokens or bank. The formal FO30 evaluator uses score threshold 0.5, mask threshold 0.5, mask-NMS IoU 0.8, 200 raw candidates, and single-instance Top-1 output. Runtime analysis separates FO Memory-driven localization from episode-local tracking. For the prompt-route comparison, the SAM3 Text Prompt and preloaded FO Memory Tokens use the same frozen 504-pixel SAM3 image backbone, fusion encoder, decoder, threshold, and post-processing. Model loading, memory-file I/O, tracker initialization, and policy inference are excluded. Five warm-up passes precede 30 CUDA-synchronized measurements. The Text Prompt route requires 48.18 ± 4.11 ms, whereas preloaded FO Memory Tokens require 43.26 ± 5.17 ms, reducing the mean by 4.92 ms (10.21%) by skipping online text encoding.

APPENDIX II

REAL-ROBOT EVALUATION PROTOCOL

The three tasks are Can Pick & Place, Box Push, and Part Assembly, covering grasp-and-place, planar contact, and local insertion. All policies are trained in a clean single-target setting. Evaluation uses an ID single-target condition, changes the background and introduces OOD-Distractors or OOD-Similar distractors, or evaluates within-type and cross-type newly registered FOs with compatible interaction geometry. OOD-Similar denotes unseen distractor instances that share the queried target’s coarse category.

Five target markers are fixed on the table, and every method uses the same cameras, object poses, marker order, termination rule, and ten- rollout protocol. ID contains only the demonstrated target. For both OOD-Distractors and OOD-Similar, the first five trials use distractor 1 and the last five use distractor 2. For within-type scaling, 3–4 similar FOs stay at fixed locations; the query switches from new FO 1 to new FO 2 after five trials. For cross-type scaling, one geometry-compatible object set is evaluated twice at each marker. Only the queried target moves across markers.

A rollout succeeds only if the robot first interacts with the queried FO and then completes the task; interacting with a wrong FO is always a failure. New-FO conditions add only FO Memory Tokens to the bank and do not add demonstrations or update FOM-SAM3, FSAE, DP, or ACT. This evaluates frozen- skill reuse for compatible geometry, not zero-shot acquisition of an arbitrary new manipulation skill. A reported success requires both correct-FO interaction and task completion; wrong-FO contact, timeout, or an unfinished task is a failure. Exact task-specific pose tolerances and timeout values are not retained in the current summary records and are therefore not reconstructed post hoc.

APPENDIX III

ADDITIONAL EXPERIMENTAL RESULTS AND OBSERVATIONS

A. Detailed FO Perception Metrics

We name the 600-image standard benchmark *FO30-Standard* (FO30-S) and the 400-image similar-object-dense stress test *FO30-Challenge* (FO30-C). These paper-facing names map to the internal evaluation profiles `standard600` and `previous400`, respectively. A similar-object-dense scene contains at least two FOs from the same predefined similar group.

TABLE VIII
COMPOSITION OF THE FO30 TEST BENCHMARKS. PRESENT AND ABSENT REFER TO EXHAUSTIVE IMAGE–FO QUERY PAIRS.

Benchmark	Images	Queries	Present	Absent	FOs / image	Similar-dense scenes
FO30-S (Standard)	600	18,000	1,268	16,732	2.11	42.8%
FO30-C (Challenge)	400	12,000	1,192	10,808	2.98	80.3%

Each test image is exhaustively paired with all 30 FO prompts, yielding 30,000 image–FO prompt tests in total: 2,460 FO-present queries and 27,540 FO-absent queries. FO30-S contains 1,268/16,732 present/absent queries, while FO30-C contains 1,192/10,808. Our method achieves GT mIoU of 93.04%/83.25% on FO-present queries, identity recall of 94.95%/86.24% and identity precision of 51.04%/56.05% over all image–FO queries, and correct-rejection rates of 93.10%/92.54% on FO-absent queries, for FO30-S/FO30-C respectively. Here, mIoU measures localization when the queried FO is present, whereas identity recall and precision measure present/absent prediction over all queries, and rejection measures correct decisions when the queried FO is absent.

Tables IX and XI provide the detailed localization, segmentation, identification, and rejection results under the same evaluator.

B. Metric Definitions

Let a positive pair contain at least one GT instance of the queried FO. GT mIoU is averaged over GT instances after Hungarian matching; an unmatched GT contributes zero, while duplicate predictions are penalized by precision/F1 rather than counted again in mIoU. F1@0.5 uses mask IoU 0.5. pmF1 averages instance-level F1 over IoU thresholds 0.50, 0.55, ..., 0.95. Joint-pass is the fraction of matcher-assigned target candidates whose score in Eq. (9) passes 0.5. Rank@1 is the fraction for which that assigned target candidate ranks first among the 200 raw candidates.

TABLE IX

GT-PRESENT LOCALIZATION AND SEGMENTATION RESULTS. ALL VALUES ARE PERCENTAGES; JOINT-PASS USES THE JOINT OBJECTNESS AND IDENTITY THRESHOLD.

Test set	Method	GT mIoU↑	pmF1↑	F1@0.5↑	Joint-pass↑	Rank@1↑
FO30-S	SAM3 Text Prompt [19]	70.18	73.57	74.00	71.85	87.85
	Text-Token Optimization [22]	84.77	81.49	82.19	86.75	90.14
	Internal Prompt Optimization [20]	68.22	73.37	73.81	69.79	87.78
	Concept Embedding Search [21]	84.85	81.53	82.31	86.91	90.54
	FO Memory Learning (Ours)	93.04	94.09	94.17	94.87	97.71
FO30-C	SAM3 Text Prompt [19]	57.46	55.89	56.23	59.06	72.48
	Text-Token Optimization [22]	68.84	62.76	63.44	70.89	75.42
	Internal Prompt Optimization [20]	53.19	56.00	56.33	54.87	71.90
	Concept Embedding Search [21]	69.63	62.95	63.69	71.73	75.17
	FO Memory Learning (Ours)	83.25	86.00	86.15	85.32	95.39

Table X gives one fixed FO diagnostic.

TABLE X

FRESH-VALIDATION DIAGNOSTIC FOR THE “COTTON SWAB BOX” FO. THE 320 SCENES CONTAIN 256 PRESENT AND 64 ABSENT CASES. VALUES ARE PERCENTAGES AND ARE SEPARATE FROM THE FORMAL FO30 TEST SETS.

Method	GT mIoU	F1@0.5	Joint-pass	Rank@1	Similar-FO FPR	Absent FPR
SAM3 Text Prompt [19]	0.00	0.00	0.00	55.47	35.55	0.00
Concept Embedding Search [21]	72.11	61.71	74.61	66.02	31.64	35.94
FO Memory Learning (Ours)	96.95	94.64	100.00	100.00	0.00	1.56

The original prompt can produce high-quality raw mask candidates for this object, but none of its matched candidates pass the joint score threshold; consequently, its GT mIoU and F1 are zero after the formal detection decision. The FO Memory Tokens learned by our method raise the target score while preserving mask quality, separate similar boxes, and retain low absent-query activation. The zero Absent FPR of the original prompt is not by itself a success: it results from near-universal rejection and is captured by the simultaneous zero recall/F1.

TABLE XI

FULL-FO IDENTIFICATION AND REJECTION RESULTS. CGF1 AND FPR VALUES ARE PERCENTAGES; IL-MCC USES ITS NATIVE SCALE. THE COUNTS COVER ALL IMAGE-FO QUERY PAIRS.

Test set	Method	cgF1↑	IL-MCC↑	Joint FPR↓	Similar-FO FPR↓	Non-similar FPR↓	TP	FP	TN	FN
FO30-S	SAM3 Text Prompt [19]	24.30	0.3304	18.96	35.46	6.03	927	3,173	13,559	341
	Text-Token Optimization [22]	35.87	0.4403	17.19	34.63	3.52	1,125	2,876	13,856	143
	Internal Prompt Optimization [20]	26.30	0.3585	15.75	29.00	5.37	906	2,636	14,096	362
	Concept Embedding Search [21]	35.82	0.4394	17.26	34.80	3.52	1,125	2,888	13,844	143
	FO Memory Learning (Ours)	62.82	0.6677	6.90	13.29	1.90	1,204	1,155	15,577	64
FO30-C	SAM3 Text Prompt [19]	20.34	0.3641	14.42	35.87	3.78	743	1,558	9,250	449
	Text-Token Optimization [22]	29.41	0.4687	13.06	34.73	2.31	891	1,411	9,397	301
	Internal Prompt Optimization [20]	20.82	0.3718	12.63	31.30	3.38	711	1,365	9,443	481
	Concept Embedding Search [21]	29.77	0.4729	13.10	34.90	2.30	900	1,416	9,392	292
	FO Memory Learning (Ours)	56.32	0.6549	7.46	17.03	2.71	1,028	806	10,002	164

APPENDIX IV

SUPPLEMENTARY VISUALIZATION PROTOCOL

A. Target Retrieval and Mask/Box Outputs

Figure 6 compares the two prompt routes on representative scenes from both FO30 test sets.

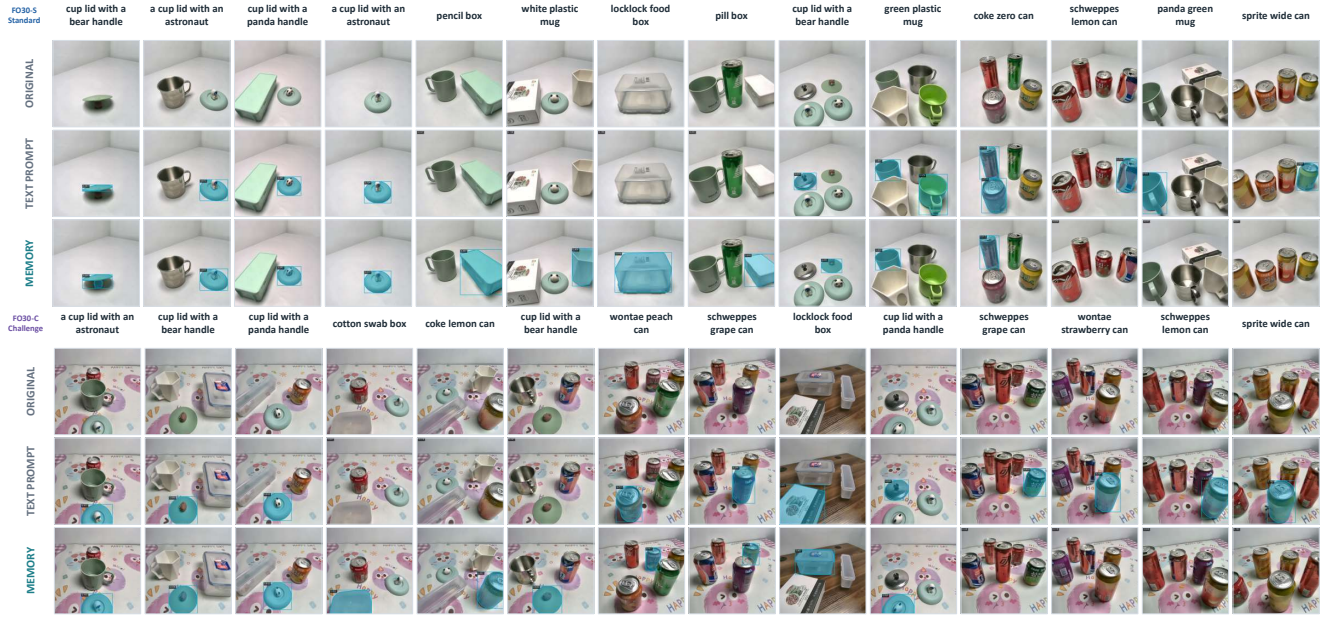


Fig. 6. **Qualitative comparison on FO30.** The top panel shows FO30-S, and the bottom panel shows FO30-C. Rows compare the original image, frozen SAM3 Text Prompt, and FOM-SAM3 with retrieved FO Memory Tokens. The examples cover target localization, Similar-FO discrimination, and absent-query rejection.

The frozen text prompt often responds to a semantically related object when the fine-grained query is ambiguous. This produces wrong-object masks in scenes with several Similar-FOs. FO Memory more often selects the queried physical identity and suppresses the other candidates. When the queried FO is absent, FOM-SAM3 returns no accepted target instead of forcing a match to the nearest Similar-FO. The same behavior is visible in the denser FO30-C scenes.

B. Offline Viewpoint-Sensitivity Analysis

We execute a fixed approach trajectory with five manually specified waypoints $P1-P5$, ordered from the initial pose toward the target can, and record one right-wrist RGB frame at each waypoint. The target pose is fixed while we repeat the trajectory under three conditions: the original can, the original can with out-of-distribution (OOD) distractors, and a replacement can from the same coarse category. This gives 15 pose-aligned frames in total. For each frame, we use the identity-conditioned SAM3 output to obtain the target box and mask, apply ROIAlign to the $256 \times 24 \times 24$ FPN feature, and retain the 256-D spatial features inside the aligned target mask. The target ROIs are visually verified because this experiment analyzes the representation after target selection rather than detector robustness.

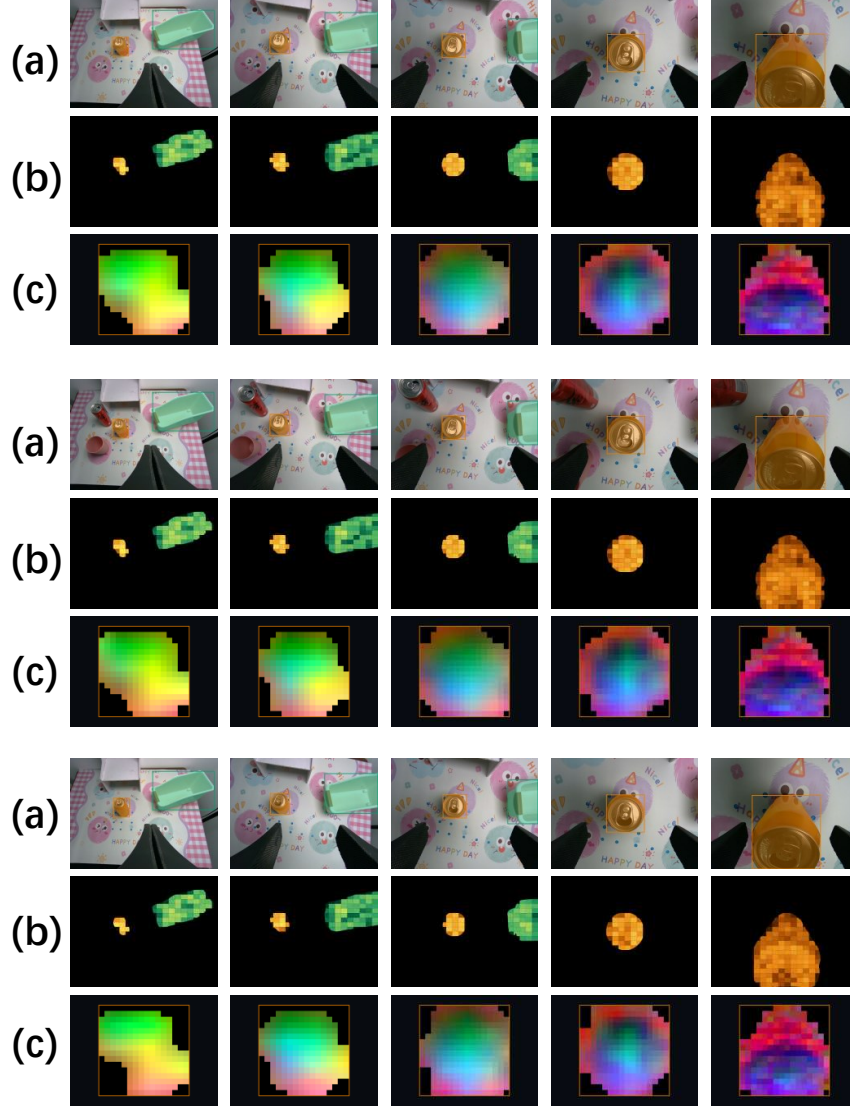


Fig. 7. **Pose-aligned target representations under three conditions.** The three blocks correspond to the original can, the original can with OOD distractors, and the replacement can; columns follow $P1-P5$. Within each block, rows show the RGB observation with predicted boxes, the mask-supported semantic features, and a visualization of the target ROI features. The box-mask operation retains target-local structure as the wrist camera approaches the object while discarding responses outside the selected target.

We jointly standardize all 3,113 valid spatial features from the 15 frames, reduce them to 50 dimensions with PCA, and fit one t-SNE embedding with perplexity 30 and random seed 42. Each small point in Fig. 8 is one mask-valid 256-D spatial feature; the outlined markers are frame centroids, and the lines indicate only the ordered $P1 \rightarrow P5$ trajectory.

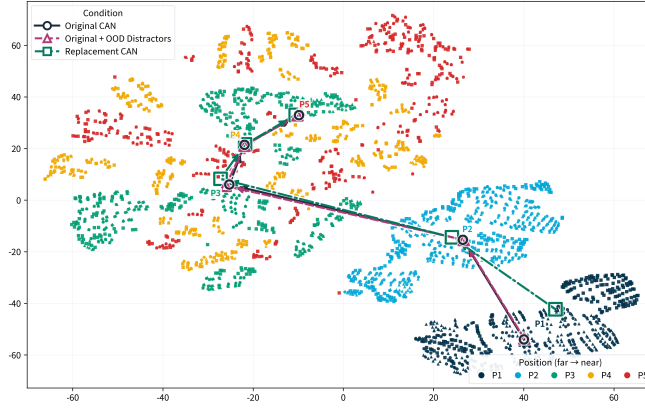


Fig. 8. **Joint t-SNE of box-mask-focused target features.** Color denotes waypoint from $P1$ (initial pose) to $P5$ (near-target pose), and marker shape denotes the three scene/instance conditions. The nearby, similarly directed centroid trajectories show that viewpoint progression is preserved across an OOD scene change and a same-category target replacement. This visualization is qualitative; t-SNE distances have no physical units.

The ordered centroid migration shows that the focused representation remains sensitive to changes in target viewpoint. At the same time, the Original and Original+OOD trajectories nearly overlap, suggesting that, once the target ROI is correctly selected, mask gating limits the influence of distractors outside the target support. The replacement can follow the same broad trajectory, especially from $P2$ to $P5$, suggesting that viewpoint change is the dominant variation along this path and that the tested within-category instance change does not disrupt the progression. The wider $P4/P5$ clouds are consistent with close views exposing more distinct can-top, wall, edge, texture, and reflection regions. These observations support the intended behavior of the FSAE front end, but they do not establish invariance or statistical significance in the original feature space.